

SPATIAL RESIDUAL PROCESSES AND BOUNDARY DETECTION

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Key words: Change-boundary statistics; regression residuals; set indexed partial sums; spatial data.

Summary: Set indexed partial sums of regression residuals obtained from the fit of a regression model to spatial data are used to define the spatial residual process. Limit processes are obtained for these partial sum sets. Statistics are derived for testing a spatial array of observations from a regression model for no change in parameters against possible alternatives involving change of parameters at unknown boundaries. These test statistics are defined in terms of set indexed partial sums, and distributional results based on properties of spatial residual processes are obtained for selected change-boundary statistics. Estimation of the locations of boundaries separating spatial sets of observations is discussed. This methodology is then applied to two sets of data.

1. Introduction

The problem of detection of parameter changes in a sequence of observations at unknown times was first considered by Shewhart (1930) who introduced the control chart approach to quality management in industrial processes. Chernoff and Zacks (1964) proposed a one-sided Bayes-type test for change of

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parameter at unknown times in sequences of random variables. For regression models, MacNeill (1978a, b) proposed a test based on cusums of residuals, and Jandhyala and MacNeill (1989, 1991) derived Bayes-type change detection statistics based on partial sums of residuals and discussed their asymptotic distributions for general regression.

Change-point methods have been extended to spatial data. Carlstein and Krishnamoorthy (1992) have considered the problem of estimating the location of an unknown boundary given that a boundary is present. They use a non-parametric approach that uses empirical distribution functions for observations assumed to be from one distribution on one side of the boundary and from a different distribution on the other side. MacNeill and Jandhyala (1993) extended some of the parametric change-point methods developed for univariate time series to the problem of testing spatial data for parameter changes at unknown location, and discussed the problem of estimating the location of boundaries between regions described by different sets of parameter values. They considered some aspects of detection and location of boundaries for linear regression.

Our interest is in testing for the presence of changes in regression parameters at unknown boundaries. For this purpose we consider a boundary to be defined by the convex hull of a set of lattice points in the d -dimensional unit cube. This set, plus the lattice points within the convex hull, is denoted by $R_{B_{n^d}}$, and the complement of $R_{B_{n^d}}$ in the unit cube is denoted by $R_{\bar{B}_{n^d}}$. The collection of such boundaries is denoted by $\mathcal{B}_{n^d}$. For large sample distribution theory and for practical applications we shall restrict attention to certain subsets of the set of convex subsets of the d -dimensional cube.

The outline of this paper is as follows. In Section 2 we review weak convergence results for set indexed partial sum processes. In Section 3 we discuss regression models for spatial data. In Section 4 we discuss how weak convergence for set indexed partial sum processes can be applied to obtain large sample theory for set indexed partial sums of regression residuals. In Section 5 statistics are derived for testing an array of observations from a regression model for no change in parameters against possible alternatives involving changes at unknown boundaries. We also discuss optimality properties of these tests. Special cases of these tests are derived in Section 6, and in Section 7 we show how large sample distributions may be obtained for selected test statistics for selected models. Estimation of the locations of boundaries is discussed in Section 8. In Section 9 we discuss the application of this methodology to the Mercer and Hall (1911) wheat-yield data and to Canadian age-period breast cancer mortality data.

2. Set indexed partial sum processes

To review some aspects of set indexed partial sums, we begin by considering an array of real valued random variables $\{\epsilon_{\mathbf{j}n} : \mathbf{j} = (j_1, j_2, \dots, j_d)\}$, $j_i \leq n$ and $n = 1, 2, \dots$ indexed by the d -dimensional positive integer lattice $\mathcal{Z}_+^d$. If we let $\mathcal{A}$ denote the collection of convex sets on the unit cube $I^d \equiv [0, 1]^d$, then for A in $\mathcal{A}$, the partial sum corresponding to A is defined to be $Z_n(A) = n^{-d/2} \sum_{\mathbf{j} \in nA} \epsilon_{\mathbf{j}n}$. Pyke (1973, 1983), Bass and Pyke (1984) and Alexander and Pyke (1986) developed a methodology for the limits of set indexed partial sum processes. If $|\cdot|$ denotes Lebesgue measure, then the Brownian process $Z = \{Z(A) : A \in \mathcal{A}\}$ is a Gaussian process with zero mean and

$$\text{Cov}(Z(A_1), Z(A_2)) = |A_1 \cap A_2|, \quad A_1, A_2 \in \mathcal{A}.$$

If $\{\epsilon_{\mathbf{j}n} : \mathbf{j} \in \mathcal{Z}_+^d\}$ are iid with mean zero and variance 1, the sequence of stochastic processes $Z_n = \{Z_n(A) : A \in \mathcal{A}\}_{n=1}^\infty$ converges weakly to the Brownian process Z (see Alexander and Pyke (1986)).

3. Regression models for spatial data

We consider n^d observations taken on a regular lattice on the unit cube, $I^d = [0, 1]^d$. The observation at the point $n^{-1}\mathbf{j} = (j_1/n, \dots, j_d/n)$ is denoted by $Y_{\mathbf{j}}$. Where no confusion should arise, the notation here and in the sequel suppresses the dependence on the sample size n . To define a general regression model, we let $f_k(\cdot, \dots, \cdot)$, $(k = 0, 1, \dots, p-1)$ be a set of d -variate non-stochastic regressor functions, and let $\{\epsilon_{\mathbf{j}} : \mathbf{j} \leq n\mathbf{1}\}$ be an array of iid real-valued random variables from the $N(0, \sigma^2)$ distribution. Then we define an array of dependent variables $\{Y_{\mathbf{j}} : \mathbf{j} \leq n\mathbf{1}\}$ on a regular lattice on the d -dimensional unit cube I^d as follows:

$$Y_{\mathbf{j}} = \sum_{k=0}^{p-1} \beta_k f_k(n^{-1}\mathbf{j}) + \epsilon_{\mathbf{j}}$$

where $\mathbf{j} = (j_1, \dots, j_d)$ is a vector of positive integers.

If we denote the vector of regression coefficients by $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_{p-1})'$, the design matrix by $\mathbf{X}$, the stacked vector of observations by $\mathbf{Y}$ and the stacked vector of noise variables by $\boldsymbol{\epsilon}$, then the model may be written in matrix form as follows:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

The least squares estimator for $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}}$ where

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

4. Limit Distribution Theory for Spatial Residual Processes

We first establish the following notation: $\mathbf{s}_d \equiv (s_1, \dots, s_d)$, and $d\mathbf{s}_d \equiv ds_1 \cdots ds_d$. The vector of regressor functions evaluated at $\mathbf{s}_d$ is then denoted by $\mathbf{f}(\mathbf{s}_d)$. Provided the squares of the regressor functions are Riemann integrable, it may be seen that the matrix $\lim_{n \rightarrow \infty} \frac{1}{n^d} (\mathbf{X}'\mathbf{X}) \equiv \mathbf{G}$ has as its $(r_1, r_2)^{th}$ component

$$\int_0^1 \cdots \int_0^1 f_{r_1}(\mathbf{s}_d) f_{r_2}(\mathbf{s}_d) d\mathbf{s}_d.$$

The inverse of $\mathbf{G}$ exists provided the regressor functions are linearly independent and square integrable; with this proviso, we define a form, $g(\mathbf{s}_d, \mathbf{t}_d)$, as follows:

$$g(\mathbf{s}_d, \mathbf{t}_d) \equiv \mathbf{f}'(\mathbf{s}_d) \mathbf{G}^{-1} \mathbf{f}(\mathbf{t}_d).$$

For any $A \in \mathcal{A}$ let $\mathbf{e}_{nA}$ denote the n^d -dimensional stacked vector that has components equal to 1 if $n^{-1}\mathbf{j} \in A$ and equal to 0 otherwise; i.e., $\mathbf{e}_{nA}$ is the indicator function for A . Then partial sums of residuals can be denoted as follows:

$$S_{gn}(A) = \mathbf{e}_{nA}' (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}') \boldsymbol{\epsilon}.$$

Hence

$$\mathbf{e}_{nA}' \mathbf{I} \boldsymbol{\epsilon} = \sum_{\mathbf{j} \in nA} \epsilon_{\mathbf{j}}.$$

If the components of $\boldsymbol{\epsilon}$ are iid with mean zero and finite variance σ^2 , then it follows from the discussion in Section 2 that:

$$\frac{1}{n^{d/2}\sigma} \mathbf{e}_{nA}' \mathbf{I} \boldsymbol{\epsilon} \Rightarrow Z(A),$$

where $\Rightarrow$ denotes weak convergence. Furthermore, it can be seen that

$$\frac{1}{n^{d/2}} \mathbf{e}_{nA}' \mathbf{X} \rightarrow \underbrace{\int \cdots \int_A \mathbf{f}(\mathbf{s}_d) d\mathbf{s}_d}_{A}.$$

Also, if $Z(\mathbf{s}_d) = Z((\mathbf{0}, \mathbf{s}])$ then it can be seen that

$$\frac{1}{n^{d/2}\sigma} \mathbf{X}'\boldsymbol{\epsilon} \Rightarrow \int_0^1 \cdots \int_0^1 \mathbf{f}(\mathbf{s}_d) dZ(\mathbf{s}_d) .$$

In summary we have the following:

Theorem If the components of $\boldsymbol{\epsilon}$ are iid with mean zero and finite variance σ^2 , and if the non-stochastic regressor functions defining the design matrix $\mathbf{X}$ are linearly independent and square integrable, then

$$\frac{1}{n^{d/2}\sigma} S_{gn}(A) \Rightarrow Z_g(A)$$

where

$$Z_g(A) = Z(A) - \underbrace{\int \cdots \int}_A \int_0^1 \cdots \int_0^1 g(\mathbf{s}_d, \mathbf{t}_d) dZ(\mathbf{t}_d) d\mathbf{s}_d .$$

The weak convergence results of Alexander and Pyke and the approach of MacNeill (1978b), which applies Theorem 5.5 of Billingsley (1968), may be used to prove this result.

It may be shown that

$$E[Z_g(A)] = Z_g(\mathbf{0}) = 0, \quad A \in \mathcal{A}$$

and that the covariance kernel for any $A, B \in \mathcal{A}$ is

$$\begin{aligned} K(A, B) &= E[Z_g(A) Z_g(B)] \\ &= |A \cap B| - \underbrace{\int \cdots \int}_A \underbrace{\int \cdots \int}_B g(\mathbf{s}_d, \mathbf{t}_d) d\mathbf{s}_d d\mathbf{t}_d . \end{aligned}$$

In the null case, the regressor functions are equal to zero; then $Y_j = \epsilon_j$, and we have $Z_g(A) = Z(A)$, where Z is a Brownian process.

If the only regressor function is a constant, then $Y_j = \beta_0 + \epsilon_j$, and we have

$$Z_g(A) = Z_0(A) = Z(A) - |A|Z(I^d),$$

where Z_0 is a Brownian bridge process.

5. Tests for changes of regression parameters at unknown boundaries

We wish to test the hypothesis of no change in the regression coefficients against certain alternatives involving changes at an unknown boundary $B \in \mathcal{B}_{n^d}$. To specify alternatives we let δ_k be the amount of change and $\omega_k(B)$ be 1 or 0 according to whether there is or is not a change at B in parameter β_k .

The model under the hypothesis of no change is given by (1), and under the supposition of a change at boundary B it is:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{H}_B\boldsymbol{\delta} + \boldsymbol{\epsilon}$$

where $\mathbf{H}_B$ is the matrix

$$\mathbf{H}_B = (\omega_0(B)f_0(n^{-1}\mathbf{j}), \dots, \omega_{p-1}(B)f_{p-1}(n^{-1}\mathbf{j}) : \mathbf{j} \leq n\mathbf{1})$$

with components replaced by 0 when $n^{-1}\mathbf{j}$ is not in $R_{B_{n^d}}$ and where $\boldsymbol{\delta} = (\delta_0, \dots, \delta_{p-1})'$ with δ_k being the amount of change in the parameter β_k ($k = 0, 1, \dots, p-1$) at the boundary B . Let the prior distribution of β_k be as follows:

$$\beta_k \sim N(0, \tau^2), \quad k = 0, 1, \dots, p-1,$$

with the components of $\boldsymbol{\beta}$ and $\boldsymbol{\epsilon}$ all distributed independently. Also, let $p(\boldsymbol{\omega}_B)$ denote the prior probability distribution of the set of unknown boundaries B for the combinations of parameters to which the changes may occur, where $\boldsymbol{\omega}_B = (\omega_0(B), \dots, \omega_{p-1}(B))'$.

The null and alternative hypotheses are

$$H_0 : \delta_0 = \dots = \delta_{p-1} = 0$$

against

$$H_1 : \delta_i \neq 0$$

for at least some i and for unknown boundary B .

Under H_0 , the p.d.f. of $\{Y_j : j \leq n\}$ is given by

$$f(\mathbf{Y}) = \left(\frac{1}{2\pi} \right)^{\left(\frac{n}{2}\right)^d} \frac{1}{|\Sigma_0|^{1/2}} \exp \left\{ -\frac{1}{2} \mathbf{Y}' \Sigma_0^{-1} \mathbf{Y} \right\},$$

where $\Sigma_0 = \mathbf{I} + \tau^2 \mathbf{X} \mathbf{X}'$.

Since Σ_0 is a positive definite matrix, there exists a full rank matrix $\mathbf{C}$ such that $\mathbf{C} \Sigma_0 \mathbf{C}' = \mathbf{I}$; i.e., $\Sigma_0^{-1} = \mathbf{C}' \mathbf{C}$. Let $\mathbf{Z} = \mathbf{C} \mathbf{Y}$. Then, under the assumption a change occurs at boundary B ,

$$(\mathbf{Z} \mid \boldsymbol{\delta}, \omega_B) \sim N(\mathbf{C} \mathbf{H}_B \boldsymbol{\delta}, \mathbf{I}).$$

Hence the p.d.f. of $(\mathbf{Z} \mid \boldsymbol{\delta})$ is

$$f(\mathbf{Z} \mid \boldsymbol{\delta}) = \sum_B p(\omega_B) f(\mathbf{Z} \mid \boldsymbol{\delta}, \omega_B)$$

where

$$f(\mathbf{Z} \mid \boldsymbol{\delta}, \omega_B) = \left(\frac{1}{2\pi} \right)^{\left(\frac{n}{2}\right)^d} \exp \left\{ -\frac{1}{2} (\mathbf{Z} - \mathbf{C} \mathbf{H}_B \boldsymbol{\delta})' (\mathbf{Z} - \mathbf{C} \mathbf{H}_B \boldsymbol{\delta}) \right\}.$$

The test statistic is defined in terms of the Laplacian of $f(\mathbf{Z} \mid \boldsymbol{\delta})$ as follows:

$$\sum_{i=0}^{p-1} \frac{\partial^2}{\partial \delta_i^2} f(\mathbf{Z} \mid \boldsymbol{\delta}) \bigg|_{\boldsymbol{\delta}=\mathbf{0}} > k f(\mathbf{Z} \mid \boldsymbol{\delta})|_{\boldsymbol{\delta}=\mathbf{0}}.$$

Then the test of H_0 versus H_1 is to reject H_0 if the statistic exceeds k where k is chosen according to the size of the test. When we evaluate the Laplacian and simplify, we obtain the following statistic:

$$Q = \sum_B p(\omega_B) \mathbf{Y}' \Sigma_0^{-1} \mathbf{H}_B \mathbf{H}_B' \Sigma_0^{-1} \mathbf{Y}.$$

Woodbury's formula, with $\tau \rightarrow \infty$, can be used to show that $\Sigma_0^{-1} = \mathbf{I} - \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'$. This Laplace-Bayesian (LpB) test statistic then may be

expressed as follows:

$$Q = \sum_B p(\omega_B) \mathbf{Y}' (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \mathbf{H}_B \mathbf{H}_B' (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \mathbf{Y}. \quad (2)$$

Provided $f(\mathbf{Z} | \delta)$ is symmetrical, a test based on the LpB statistic Q is locally most mean powerful (LMMP). This optimality property, established by SenGupta and Vermeire (1986), is a generalization to the multiparameter case of locally best unbiased tests.

Note that the LpB statistic Q is defined in terms of squares of set indexed partial sums of residuals.

The test statistic Q as defined in (2) tests for the possibility of change in all the regression parameters. If one wishes to test for change in only a subset of the regression parameters, then the appropriate LpB statistic can be obtained from (2) by including in $\mathbf{H}_B$ only those columns corresponding to the parameters of interest.

6. Test statistics for change of mean at unknown rectangular boundaries

In this section we illustrate the results of the previous section by considering several simple cases. In particular, we consider a two-dimensional array of n^2 normally distributed observations $Y_{ij}(i, j = 1, \dots, n)$ and test for the presence of a change in mean at an unknown rectangular boundary. This is a special case of the regression model and the unknown lattice boundaries considered in Sections 1 and 3.

First we consider the collection of boundaries consisting of the rectangles whose left lower corner is $\mathbf{0} = (0, 0)$ and whose sides are parallel to the sides of the unit square. Such a rectangular boundary requires only another right upper corner $n^{-1}\mathbf{k} = (k_1/n, k_2/n)$ to fix the entire boundary. The boundary

is denoted by $B_{\mathbf{k}}$, and the lattice points contained by the rectangle, denoted by $R_{B_{\mathbf{k}}}$, are the points $n^{-1}\mathbf{1} \leq n^{-1}\mathbf{j} \leq n^{-1}\mathbf{k}$, where $\mathbf{j} = (j_1, j_2)$.

Using the results of Section 3, we find the LpB statistic to be:

$$Q_{1n} = \sum_{k_1=1}^n \sum_{k_2=1}^n p(\omega_{\mathbf{k}}) \omega_{\mathbf{k}} \left\{ \sum_{j_1=1}^{k_1} \sum_{j_2=1}^{k_2} (Y_{j_1 j_2} - \hat{\beta}_0) \right\}^2.$$

A rectangular boundary with sides parallel to those of the entire space, but located at any position within the unit square, requires two points $n^{-1}\mathbf{k} = (k_1/n, k_2/n)$ and $n^{-1}\mathbf{r} = (r_1/n, r_2/n)$ ($\mathbf{k} \leq \mathbf{r}$) to entirely fix the boundary. Such a boundary is denoted by $B_{\mathbf{rk}}$.

Again, using the results of the previous section, we obtain the LpB statistic:

$$Q_{2n} = \sum_{k_1=1}^n \sum_{k_2=1}^n \sum_{r_1=k_1}^n \sum_{r_2=k_2}^n p(\omega_{\mathbf{rk}}) \omega_{\mathbf{rk}} \left\{ \sum_{j_1=k_1}^{r_1} \sum_{j_2=k_2}^{r_2} (Y_{j_1 j_2} - \hat{\beta}_0) \right\}^2.$$

It is straightforward to extend these results to more general regression models and to the d-dimensional unknown rectangular solid boundary case. Other collections of boundaries may be considered; these result in test statistics consisting of other functionals of the partial sums of residuals.

7. Probability distributions for change – boundary statistics for selected models

The LpB statistic for detecting change of intercept in a regression model for the set of boundaries $\mathcal{A}$ as given by (2) and with a uniform prior distribution on the boundaries is

$$Q_g = \sum_A \left(\sum_{\mathbf{j} \in nA} r_{\mathbf{j}} \right)^2.$$

Under H_0 ,

$$\frac{1}{n^{2d}\sigma^2}Q_g \rightarrow \underbrace{\int_0^1 \cdots \int_0^1}_d Z_g^2(A)dA = \int_{I^d} Z_g^2(A)dA, \quad A \in \mathcal{A}.$$

Example For detecting a change in the intercept of a regression model for two-dimensional data and boundaries B_k with one vertex fixed at $\mathbf{0}$ the LpB statistic is

$$Q_{g1n} = \sum_{l=1}^n \sum_{k=1}^n \left(\sum_{i=1}^l \sum_{j=1}^k r_{ij} \right)^2.$$

Under H_0 ,

$$\frac{1}{n^4\sigma^2}Q_{g1n} \rightarrow \int_0^1 \int_0^1 Z_g^2(t_1, t_2)dt_1dt_2.$$

Large sample quantiles for Cramér-von Mises statistics of this type are obtained through evaluation of the distribution of the relevant stochastic integrals. MacNeill (1978a), Jandhyala and MacNeill (1991) and Jandhyala and Minogue (1993) developed several approaches for computing quantiles for the Cramér von-Mises type stochastic integrals that arise in the large sample theory for change-point statistics. However, there are difficulties, mainly involving multiplicities of the eigenvalues of the Karhunen-Loeve expansion of the limit process, in extending these approaches to higher dimensions.

In order to solve

$$\alpha = P \left(\int_{I^d} Z_g^2(A)dA \leq q \right)$$

for α for given $q > 0$, and conversely q for given probability level α , we consider the method introduced by Imhof (1961) and applied to change-point statistics by Eastwood (1993) to obtain approximate distributions as follows:

$$P \left(\int_{I^d} Z_g^2(A)dA \leq q \right) \simeq P \left(\chi_h^2 \leq q' \right),$$

where χ_h^2 denotes a central chi-square random variable with h degrees of freedom, where q and q' are related by the equation $q' = (q - c_1)\sqrt{\frac{h}{c_2}} + h$, where $h = \frac{c_3^3}{c_2^3}$ and where c_i ($i = 1, 2, 3$) is the i^{th} cumulant.

We apply these methods to obtain quantiles for change boundary statistics for the following models: when the model is constant,

$$Y_{ij} = \beta_0 + \epsilon_{ij} ;$$

when the model is linear,

$$Y_{ij} = \beta_0 + \beta_1(i/n) + \beta_2(j/n) + \epsilon_{ij} ;$$

and when the model is quadratic,

$$Y_{ij} = \beta_0 + \beta_1(i/n) + \beta_2(j/n) + \beta_{11}(i/n)^2 + \beta_{22}(j/n)^2 + \beta_{12}(ij/n^2) + \epsilon_{ij} .$$

Table 1 contains selected quantiles for the distribution of

$$\int_0^1 \int_0^1 Z_g^2(t_1, t_2) dt_1 dt_2 .$$

Table 1. Selected quantiles for selected change-boundary statistics.

α	null	constant	linear	quadratic
.5	.16101	.10547	.06608	.04383
.9	.53805	.25966	.11503	.04560
.95	.72413	.33111	.12742	.05435
.975	.91678	.40397	.14371	.07246
.99	1.17825	.50169	.16461	.10799
.995	1.37971	.57639	.18007	.14101

8. Estimating the location of a change – boundary

The problem of estimating the location of a change-boundary, given that one is present, was considered in a non-parametric setting by Carlstein and Krishnamoorthy (1992). Their approach uses empirical distribution functions

for observations assumed to be from one distribution on one side of the boundary and from a different distribution on the other side. The selected boundary maximizes one of several proposed criteria.

MacNeill and Jandhyala (1993) proposed ad hoc local procedures for estimating the locations of change-boundaries in spatial data. These methods reduce substantially the amount of computing required by estimation procedures that make few or no restrictions on the size of the collection of possible boundaries. Computing becomes an important issue as the grid size in the lattice shrinks.

Esterby and El-Shaarawi (1981) discussed maximum marginal and conditional likelihood methods for estimating the location of change-points in regression models. If an unknown boundary B requires only a finite number of points to fix the entire boundary (e.g., the rectangular boundaries discussed in Section 3), we can apply these methods to estimate the locations of boundaries as follows. Let R_B be a collection of points, $n^{-1}\mathbf{j}$, on a grid contained in a region within or on one side of boundary B which can be fixed by a finite number of points and let $R_{\bar{B}}$ be the complementary set. Let $r_{\mathbf{j}}$ be the residual associated with the point $n^{-1}\mathbf{j}$. Then let

$$S_{R_B}^2 = \sum_{(n^{-1}\mathbf{j}) \in R_B} r_{\mathbf{j}}^2,$$

and

$$S_{R_{\bar{B}}}^2 = \sum_{(n^{-1}\mathbf{j}) \in R_{\bar{B}}} r_{\mathbf{j}}^2.$$

We estimate as the boundary B^* that boundary B which minimizes,

$$S_{R_B R_{\bar{B}}}^2 = S_{R_B}^2 + S_{R_{\bar{B}}}^2.$$

That is

$$S_{B^* \bar{B}^*}^2 = \min_B S_{R_B R_{\bar{B}}}^2.$$

9. Applications

We consider the wheat-yield data compiled by Mercer and Hall (1911) and discussed by Cressie (1993). The data are yields of grain (in pounds) for a 20×25 lattice of plots with 20 rows of plots running east to west and 25 columns of plots running north to south; Figure 1 presents the data.

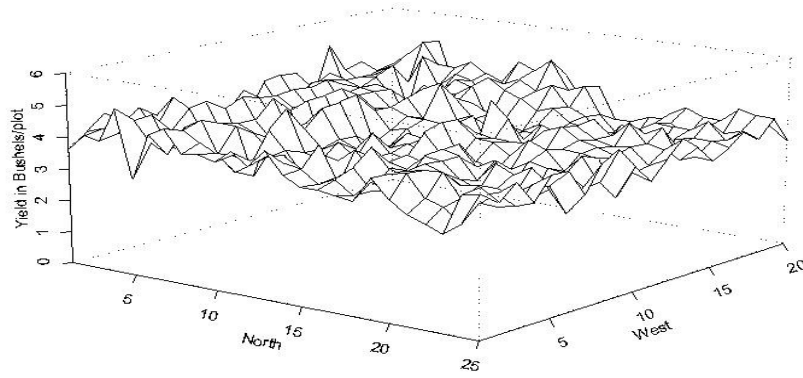


Figure 1. Mercer and Hall wheat-yield data.

We test the data for change of mean level at unknown boundary. The collection of boundaries we consider first are those defined by the rectangles with sides parallel to those of the field and with one corner fixed in the north-west corner of the field. These are examples of the first set of rectangular boundaries discussed in Section 6. An LpB statistic for detecting change in mean for spatial data of this type is analogous to Q_{1n} in Section 7 and is defined as follows:

$$Q_{1cn} = \sum_{l=1}^n \sum_{k=1}^{[cn]} \left(\sum_{i=1}^l \sum_{j=1}^k (Y_{ij} - \bar{Y}) \right)^2 ,$$

where $c > 0$ is a constant and $[cn]$ is the largest integer in cn ; this choice of lattice permits the use of statistics based on rectangular data sets which retain the same large sample properties as those of Section 7. The hypotheses we are testing are as follows:

$$H_0 : E(Y_{ij}) = \mu_0, \text{ for all } (i, j)$$

versus

$$\begin{aligned} H_1 E(y_{ij}) &= \mu_0 && \text{for all } (i, j) \leq (l, k) \\ &= \mu_1 \neq \mu_0 && \text{otherwise} \end{aligned}$$

Under H_0 , it can be shown that

$$\frac{1}{c^2 n^4 \sigma^2} Q_{1cn} \rightarrow \int_0^1 \int_0^1 Z_g^2(t_1, t_2) dt_1 dt_2 .$$

If σ^2 is not known, then an estimator that is consistent under both H_0 and H_1 can be obtained using differences in a regular grid. For example, if

$$d_{ij} = Y_{ij} - \frac{1}{4}(Y_{i-1,j} + Y_{i,j-1} + Y_{i+1,j} + Y_{i,j+1}) ,$$

then $E(d_{ij}^2) = \frac{5}{4}\sigma^2$. Hence we estimate σ^2 by $\tilde{\sigma}^2$ where

$$\tilde{\sigma}^2 = \frac{4}{5} \frac{1}{(n-2)([cn]-2)} \sum_{i=2}^{n-1} \sum_{j=2}^{[cn]-1} d_{ij}^2 .$$

For the Mercer and Hall wheat-yield data, $n = 20$, $c = 5/4$, $\tilde{\sigma}^2 = 0.1194$ and we obtain $\frac{Q_{1cn}}{c^2 n^4 \tilde{\sigma}^2} = 3.4033$. Since the 0.995 quantile for $\frac{Q_{1cn}}{c^2 n^4 \tilde{\sigma}^2}$ is 0.57639 (see Table 1), a change in mean level is detected.

To estimate the location of the boundary we adapt the methods of Section 8. Let R_B be a collection of points, $(i/n, j/[cn])$, on a grid contained in a region within or on one side of boundary B which is fixed by the points $\mathbf{1}$ and $\mathbf{k}$, and let $R_{\bar{B}}$ be the complementary set. Let r_{ij} be the residual associated with the point $(i/n, j/[cn])$. For the Mercer and Hall wheat-yield data, the unknown change-boundary is estimated to be B^* where R_{B^*} is defined as

follows: $R_{B^*} = \{(i, j) : i = 1, \dots, 20, j = 1, \dots, 11\}$.

Figure 2 is a graph of $-S_{R_B R_{\bar{B}}}^2$, which defines the marginal likelihood of the boundary location.

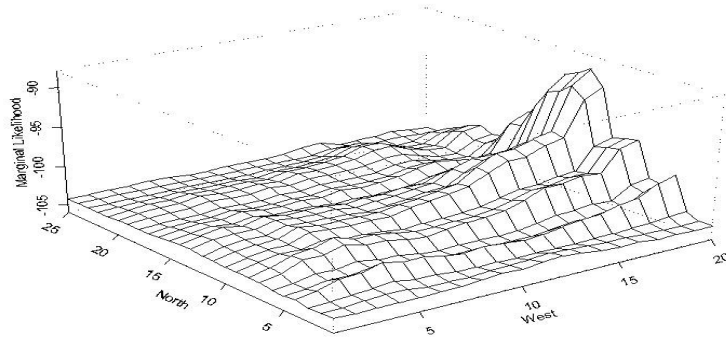


Figure 2. Marginal likelihood for the boundary location in the Mercer and Hall data.

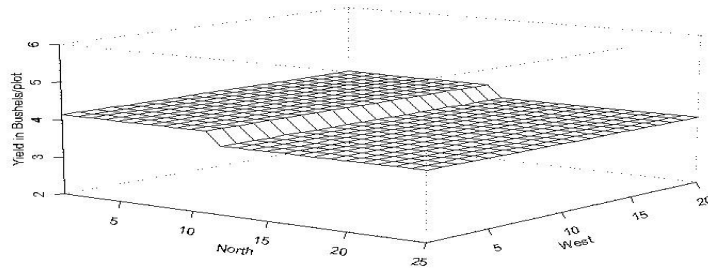


Figure 3. Fitted model for a change-boundary based on B^* for the Mercer and Hall wheat-yield data.

The estimate of the mean for the set R_{B^*} is $\hat{\mu}_0 = 4.140$ and for the set $R_{\bar{B}^*}$ it is $\hat{\mu}_1 = 3.795$. The fitted model with the parameter change appears in Figure 3.

We also consider the collection of boundaries defined by the rectangles with sides parallel to the sides of the field but located at any position. These are examples of the second set of rectangular boundaries discussed in Section 6. For the Mercer and Hall wheat-yield data, the unknown change-boundary is estimated to be A^* where $R_{A^*} = \{(i, j) : i = 1, \dots, 20, j = 3, \dots, 11\}$. The estimate of the mean for the set R_{A^*} is 4.189 and for the set $R_{\bar{A}^*}$ is 3.813. The fitted model with change-boundary A^* appears in Figure 4.

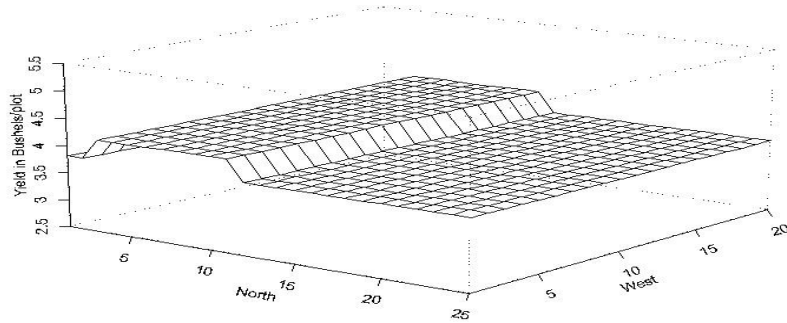


Figure 4. Fitted model for a change-boundary based on A^* for the Mercer and Hall wheat-yield data.

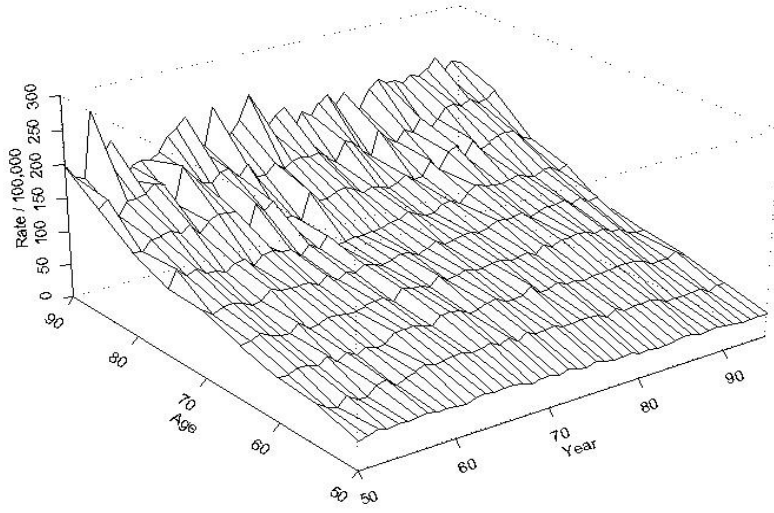


Figure 5. Canadian post-menopausal breast cancer mortality rates.

The next example concerns Canadian breast cancer mortality rates. Figure 5 presents post-menopausal rates for the years 1950 – 1990 for five year age groups beginning at age 50. We first fitted the plane

$$Y_{ij} = \beta_0 + \beta_1(i/n) + \beta_1(j/n) + \epsilon_{ij}$$

to the data and computed the residuals. Then, assuming boundaries B_k with one vertex fixed at $\mathbf{0}$, we computed the value of Q_{1cn} using $n = 8, c = 41/8$. With $\tilde{\sigma}^2 = 9.37$ (based on second differences), we obtain $\frac{Q_{1cn}}{c^2 n^4 \tilde{\sigma}^2} = 0.99$. Since the 0.995 quantile for $\frac{Q_{1cn}}{c^2 n^4 \tilde{\sigma}^2}$ is 0.18007 (see Table 1), a change in intercept is detected.

To estimate the location of the boundary we again adapt the methods of Section 8. For the breast cancer mortality data, the unknown change-boundary

is estimated to be B^{**} where $R_{B^{**}}$ is defined as follows:

$$R_{B^{**}} = \{(i, j) : i = 1, \dots, 41, j = 1, \dots, 4\}.$$

Figure 6 presents a graph of the fitted change-boundary model based on B^{**} .

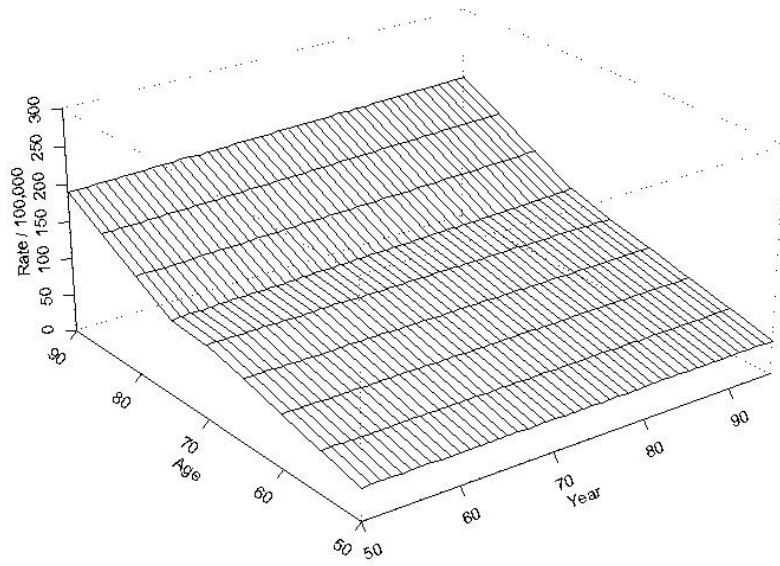


Figure 6. The fitted change-boundary model based on B^{**} for Canadian post-menopausal breast cancer mortality rates.

Parameter estimates for the set $R_{B^{**}}$ are: $\hat{\beta}_{01} = -96.5012$, $\hat{\beta}_{11} = 2.7963$, $\hat{\beta}_{21} = 0.0433$; and for the set $R_{\bar{B}^{**}}$ they are: $\hat{\beta}_{02} = -333.3857$, $\hat{\beta}_{12} = 5.8025$, $\hat{\beta}_{22} = 0.3358$.

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